

$$\int \frac{\sqrt{4x^2 - 25}}{x^3} dx = \int \frac{5 \tan \theta \cdot \cancel{\frac{5}{2}} \sec \theta \tan \theta d\theta}{\frac{5}{2} \cdot \frac{5}{2} \cdot \cancel{\frac{5}{2}} \sec^3 \theta}$$

$$2x = 5 \sec \theta$$

$$2dx = 5 \sec \theta \tan \theta d\theta$$

$$\sqrt{25 \sec^2 \theta - 25} = \sqrt{25(\sec^2 \theta - 1)} = 5 \tan \theta$$

$$= \int \frac{5 \tan^2 \theta}{\sec^2 \theta} \cdot \frac{2 \cdot 2}{5 \cdot 5} d\theta$$

$$= \frac{4}{5} \int \frac{\tan^2 \theta}{\sec^2 \theta} d\theta = \frac{4}{5} \int \frac{\sin^2 \theta}{\cos^2 \theta} d\theta$$

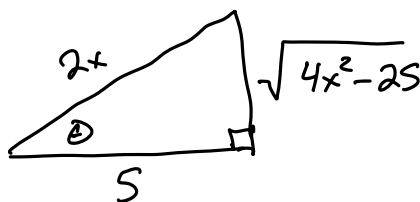
$$= \frac{4}{5} \int \sin^2 \theta d\theta = \frac{4}{5} \frac{1}{2} \int 1 - \cos 2\theta d\theta = \frac{2}{5} \left[\theta - \frac{1}{2} \sin(2\theta) \right] + C$$

$$= \frac{2}{5} \left[\theta - \sin \theta \cos \theta \right] + C$$

$$= \frac{2}{5} \left[\operatorname{arcsin}\left(\frac{2x}{5}\right) - \frac{\sqrt{4x^2 - 25} \cdot 5}{(2x)^2} \right] + C$$

$$= \frac{2}{5} \left[\arccos\left(\frac{5}{2x}\right) - \frac{5\sqrt{4x^2 - 25}}{4x^2} \right] + C$$

$$\sec \theta = \frac{2x}{5} = \frac{\text{hyp}}{\text{adj}}$$



$$\theta = \arcsin\left(\frac{2x}{5}\right)$$

$$\cos \theta = \frac{5}{2x}$$

$$2. \int \frac{dx}{\sqrt{x^2+4x+8}}$$

$$(x^2+4x+\underline{4})+8-\underline{4}$$

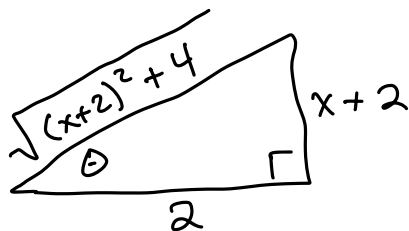
$$(x+2)^2+4$$

$$\int \frac{dx}{\sqrt{(x+2)^2+4}}$$

$$\begin{aligned} x+2 &= 2 \tan \theta \\ dx &= 2 \sec^2 \theta d\theta \\ \sqrt{4 \tan^2 \theta + 4} &= 2 \sec \theta \end{aligned}$$

$$= \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$\tan \theta = \frac{x+2}{2}$$



$$= \ln \left| \frac{\sqrt{(x+2)^2+4}}{2} + \frac{x+2}{2} \right| + C$$

OR

$$= \ln |\sqrt{(x+2)^2+4} + x+2| + C$$

2.0.2 Example:

Write out the partial fraction decomposition for

$$\frac{x^2 + 2x + 1}{x(x+2)^2(x+5)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{x+5}$$

2.0.3 Example:

$$\int \frac{4x}{(x-1)^2(x+1)} dx \left[\frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} \right] (x-1)^2(x+1)$$

$$4x = A(x^2 - 1) + B(x+1) + C(x-1)^2$$

To get A, look at coeff. of x^2

plug in $x=1$	plug in $x=-1$
$4 = B(2)$	$-4 = C(-1-1)^2$
$2 = B$	$-4 = 4C$
	$-1 = C$

$$\begin{aligned} 0 &= A + C \\ 0 &= A - 1 \\ A &= 1 \end{aligned}$$

$$\int \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} dx = \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + C$$

$$\begin{aligned} u &= x-1 \\ du &= dx \\ \int \frac{2}{u^2} du \\ \int 2u^{-2} du \\ \frac{2u^{-1}}{-1} \end{aligned}$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

2.0.4 Example:

$$\int \frac{3x+4}{x^2+9} dx = \int \frac{3x}{x^2+9} dx + \int \frac{4}{x^2+9} dx$$

$\uparrow$
 not factorable

$u = x^2 + 9$
 $du = 2x dx$
 $\int \frac{3x}{u} \frac{du}{2x}$
 $\frac{3}{2} \int \frac{1}{u} du$

$4 \int \frac{1}{x^2+9} dx$
 $= \frac{4}{3} \arctan\left(\frac{x}{3}\right)$

$= \frac{3}{2} \ln(x^2+9) + \frac{4}{3} \arctan\left(\frac{x}{3}\right) + C$

3 Case 3: Denominator has an irreducible quadratic factor

An irreducible quadratic factor may look like $x^2 + 4$ or $x^2 + 7x + 4$.

For each irreducible quadratic factor $ax^2 + bx + c$ in the denominator, include $\frac{Ax+B}{ax^2+bx+c}$ in the partial fraction decomposition.

3.0.1 Example:

Write out the partial fraction decomposition for

$$\frac{\text{whatever}}{(x^2+4)(x+1)^2(x-2)} = \frac{A x + B}{x^2+4} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{x-2}$$

3.0.2 Example:

$$\int \frac{4x^2 + 8x}{(x^2 + 4)(x^2 - 4)} dx \quad \left[\frac{4x^2 + 8x}{(x^2 + 4)(x-2)(x+2)} = \frac{Ax+B}{x^2+4} + \frac{C}{x-2} + \frac{D}{x+2} \right] (x^2+4)(x-2)(x+2)$$

$$4x^2 + 8x = \overset{Ax^3}{(Ax+B)} \overset{B(-4)}{(x-2)(x+2)} + \overset{Cx^3}{C} \overset{C \cdot 8}{(x^2+4)(x+2)} + \overset{Dx^3}{D} \overset{D(-8)}{(x^2+4)(x-2)}$$

plug in $x=2$

$$4 \cdot 4 + 8 \cdot 2 = C(4+4)(4)$$

$$32 = 32C$$

$$1 = C$$

plug in $x=-2$

$$4 \cdot 4 + 8(-2)$$

$$0 = D(4+4)(-4)$$

$$0 = D$$

x^3

$$0 = A + C + D$$

$$0 = A + 1 + 0$$

$$-1 = A$$

Constant

$$0 = -4B + 8C - 8D$$

$$0 = -4B + 8$$

$$4B = 8 \quad B = 2$$

$$\int \frac{-x+2}{x^2+4} + \frac{1}{x-2} dx$$